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Department Of Physics

# Numerical investigation of propeller noise reduction

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Submitted in partial fulfillment of the requirements for the degree of
Master of Physics
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# Certificate

This is to certify that the project titled **Numerical Investigation of Propeller Noise Reduction** has been conducted by **Diya K D** for the partial fulfillment of M.Sc. Degree in Physics, as stipulated by St Joseph's University was performed under my guidance. I certify that the project is her bona fide work and has been performed to my satisfaction.

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# Declaration

I, **Diya K D** declare that to the best of my conscience, the project entitled **Numerical Investigation of Propeller Noise Reduction** was performed by me, under the guidance of **Dr. Jayahar Sivasubramanian** The work has not been submitted in substance before for any degree.

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# Acknowledgments

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20th November, 2023.

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## LIST OF ABBREVIATIONS:

1. ANC: Active Noise Cancellation
2. ML: Machine learning
3. FFT: Fast Fourier Transform
4. ISR: Intelligence, Surveillance, and Reconnaissance
5. PA-ANC: Pitch-Aligned Active Noise Cancellation.
6. LMS: Least Mean Squares
7. RLS: Recursive Least Squares
8. FIR FILTERS: Finite Impulse Response (FIR) filters
9. NN: Neutral Noise
10. UAV: Unmanned Arial Vehicle
11. DSP: Digital Signal Processor

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# 1. INTRODUCTION

Drones are everywhere these days, doing cool things like delivering packages and taking awesome pictures. But there's a downside - they can be pretty noisy, mainly because of their spinning propellers. This thesis dives into the world of drone noise and aims to make it quieter by using computer simulations and Active Noise Cancellation (ANC).

The concept is simple: we think if we carefully listen to the noise the drone makes, we can create a special sound that cancels it out, making the drone much quieter. To do this, we'll study the pattern of the drone noise and come up with a kind of "anti-noise" signal. It's like having a secret weapon against drone noise! By doing this, we hope to make drones more friendly to our ears and the environment.

But that's not all – we'll also figure out how to make sure this anti-noise signal lines up perfectly with the drone noise. It's a bit like a musical tune-up to make sure everything works together smoothly. By doing this, we hope to make drones way less noisy and way more neighbor-friendly. Our exploration goes beyond noise reduction; it's about enhancing drones for everyone's benefit, contributing to advancements in drone technology.

This study is not just about reducing noise; it's about making drones better for everyone. The ultimate objective is to contribute to the refinement of drone technologies, rendering them not only quieter but also more enjoyable for users and less intrusive to the surrounding environment.

## 1.1 Noise Pollution

Noise pollution is like having too much of those annoying or disruptive sounds around us. It comes from things like traffic, construction, and all sorts of loud activities. The problem is, that it's not just irritating; it can mess with our health and well-being.

Imagine living with constant loud noises – it can stress us out, mess up our sleep, and even lead to hearing problems. And it's not just us; it affects animals and the environment too. Birds might change their behavior, and the whole natural balance can get messed up.

Spy drones are unmanned aerial vehicles that are used for intelligence, surveillance, and reconnaissance (ISR) purposes. They are equipped with cameras, sensors, and other surveillance equipment that allow them to gather information about enemy forces, troop movements, and strategic assets. Spy drones can operate undetected because they have quiet propellers, which reduces the risk of detection by enemy forces. This makes them invaluable assets for militaries that need to gather intelligence without putting human pilots at risk.

Now, imagine adding drones to the mix, especially when they're out there delivering packages. Those powerful rotor sounds from the drones not only bother us in cities but also disrupt the peace in places that were quiet before, like residential and rural areas.

Interestingly, studies say that people find drone noise even more bothersome than the usual sounds from cars. Drones are like the new, unnecessary noise-makers in our lives.

Even in the film world, drones create a headache. Their propeller noise messes up live recordings, so moviemakers often have to replace the original sound with something else.

People are working on ways to tackle this noise mess, using things like noise cancellation technology and rules about how much noise is okay. But with more and more drones buzzing around, we need a smart plan to deal with the noise they bring. Balancing the challenges of city noise and drone sounds is essential to keep our living spaces peaceful and happy.

## 1.2 Noise Cancellation

Noise cancellation, also known as active noise control, is a sophisticated audio processing technique designed to reduce or eliminate unwanted ambient sounds. Unlike passive noise reduction, which relies on physical barriers to block sound, active noise cancellation employs advanced electronic systems to neutralize external disturbances.

The method includes:
1. Recording a propeller noise using a microphone.
2. Analyzing the sound wave.
3. Creating the anti-noise signal of the analyzed sound wave.
4. Cancellation of the sound wave when the noise signal and anti-noise signal interfere with each other.

Active noise cancellation performs well in low frequency ranges while passive noise cancellation performs well in higher frequency ranges.

## 1.3 Problem Statement:

The increasing abundance of drones has led to a dangerous escalation in noise pollution. Current noise reduction methods, including alterations to propeller shapes and the implementation of active noise cancellation techniques, can be costly and may adversely affect drone performance.

Therefore, there is a need for a solution that actively cancels drone noise using an external anti-noise signal without imposing financial burdens or performance constraints.

## 1.4 Thesis Contribution:

This thesis hypothesizes that effective drone noise reduction can be achieved through Pitch-Aligned Active Noise Cancellation (PA-ANC). The PA-ANC system leverages the principle of beats, utilizing oscillations in amplitude resulting from the interaction of two signals with different frequencies. By manipulating the dominant frequency or pitch of the anti-noise signal, the PA-ANC system precisely aligns its phase with the drone noise signal, achieving maximum noise reduction when the phase difference reaches 180 degrees. Also, I used FFT and took inverse FFT and found the anti-noise signal.

# 2. LITERATURE REVIEW

Signal processing algorithms such as Least Mean Squares (LMS) and Recursive Least Squares (RLS) are essential tools for tuning filters in real time, ensuring that the output sound is precise and clean. These dynamic filters are especially useful for applications where the input signal characteristics may vary over time, such as in noise-canceling systems.

## 2.1 Least Mean Squares (LMS):

The Least Mean Squares (LMS) algorithm is a widely used method for adaptive filtering. It operates by iteratively adjusting the filter coefficients to minimize the mean square error between the desired signal and the actual signal. In the context of noise cancellation, LMS algorithms aim to mimic a desired filter by continuously updating coefficients to reduce the error signal, which represents the difference between the desired and actual signals. LMS filters are computationally efficient and are often suitable for real-time applications that require online adjustments.

## 2.2 Recursive Least Squares (RLS):

The Recursive Least Squares (RLS) algorithm is different from the Least Mean Squares (LMS) algorithm in that it recursively updates its filter coefficients to minimize a weighted linear least squares cost function as new data becomes available. Although RLS is more computationally demanding than LMS, it offers faster convergence and better tracking of time-varying signals. This algorithm is especially useful in situations where there are sudden changes in the input signals' characteristics.

## 2.3 Adaptability and FIR Filters:

LMS and RLS are both types of Finite Impulse Response (FIR) filters. Their adaptability lies in their capability to modify the coefficients in real time, which makes them suitable for tasks such as noise cancellation. Nevertheless, it is crucial to emphasize that these adaptive filters, including LMS and RLS, may require some time to adjust to variations in the environment or input signal. It's worth noting that the adaptive filters begin to adjust to the noise pattern after processing a specific number of samples.

## 2.4 Neutral Noise in ANC:

Active Noise Cancellation (ANC) involves using a "Neutral Noise" model that can predict the values of noise samples at the microphone's position. This predictive ability is essential for effectively canceling out noise in real-time. Adaptive filters, such as LMS and RLS, are used to continuously update the coefficients and adapt to changing environmental noise. This ensures optimal cancellation performance.

In summary, adaptive LMS and RLS algorithms can adjust their coefficients in response to changing input signals, making them valuable for applications like noise cancellation where environmental noise conditions can vary.

## 2.5 Propellers

### 2.5.1 Types of Propellers:

Propellers are essential components of many vehicles, including aircraft, boats, and ships. They come in various types, but generally, they can be divided into two main categories: fixed-pitch propellers and variable-pitch propellers.

Fixed-pitch propellers have blades that are fixed at a certain angle, and this angle cannot be changed during operation. This means that the amount of thrust generated by the propeller remains constant, making it ideal for applications where a constant speed is needed, such as in small aircraft and boats.

On the other hand, variable-pitch propellers have blades that can be rotated to change their angle of attack. This allows the propeller to generate more or less thrust as needed, making it suitable for applications where variable speed is required, such as in large aircraft and ships. The angle of attack of the blades can be adjusted manually or automatically, depending on the design.

Furthermore, variable-pitch propellers are highly efficient at different altitudes, as they can adjust to changes in air density. This makes them more effective than fixed-pitch propellers, which tend to be less efficient at higher altitudes. However, it should be noted that variable-pitch propellers are also more complex and expensive than fixed-pitch propellers, and therefore require more maintenance.

### 2.5.2 Propeller Noise

Propeller noise is the sound generated by the interaction of propeller blades with the air. The propeller noise is caused by three main mechanisms:

- Blade slapping: This happens when the blades strike the air and are most common at low speeds, resulting in a high-frequency noise.

- Vortex shedding: This noise is produced by the vortices created at the tips of the blades, which is most common at high speeds and results in a low-frequency noise.

- Broadband noise: This type of noise is caused by factors such as turbulence and cavitation, and is typically present at all speeds as a constant background noise.

### 2.5.3 Propeller Design

The design process for propellers must consider thrust, efficiency, and noise. The following are some of the key design considerations for propellers:

- Blade shape: The shape of the propeller blades is a critical factor in determining the propeller's performance. The blade shape must be designed to produce the desired thrust and efficiency while minimizing noise.
- Blade number: The number of blades on a propeller is another important factor in determining its performance, where more blades can generate more thrust, but they also generate more noise.
- Blade material: The material of the propeller blades is also important. Propeller blades are typically made from aluminum or composite materials. Aluminum blades are lighter and less expensive, but composite blades are stronger and more durable.

### 2.5.4 Reducing Propeller Noise

There are several ways to reduce propeller noise. Some of the most common methods include:

- Using quieter propeller designs: This can be done by using blades with a more favorable shape, using fewer blades, and using quieter blade materials.
- Using noise-reducing coatings: These coatings can be applied to the propeller blades to absorb sound waves and reduce noise levels.
- Using noise-reducing enclosures: Ducting and Shrouding: these enclosures can be placed around the propeller to block sound waves from reaching the listener.

Propellers play a vital role in the functioning of various types of vehicles such as aircraft and marine vehicles. They are responsible for converting the power generated by the engine into thrust, which is necessary for the movement of the vehicle through air or water. However, propellers can also be a significant source of noise. Therefore, it is crucial to design them carefully to minimize noise.

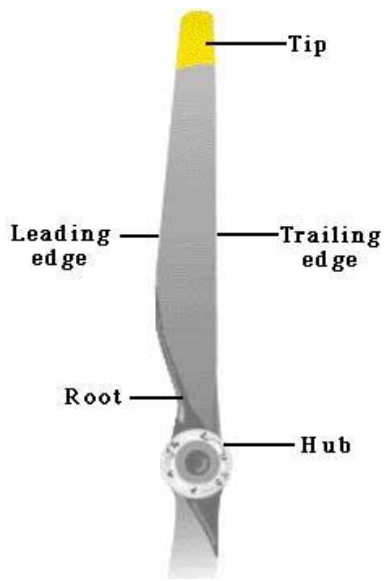


*Fig 1: propeller design (Image from Wikipedia)*

## 2.6 Interference

In the world of waves—think sound, light, or water—when two or more waves meet, they interact in interesting ways called interference. Imagine two friends jumping on a trampoline. When they land together (in phase), their jumps add up, making a higher bounce (constructive interference). But if one friend lands as the other takes off (out-of-phase), they might cancel each other out, resulting in no bounce at all (destructive interference).

This idea of waves interacting has a big impact across different areas:

### 2.6.1 Constructive Interference:

When waves align, they team up, getting stronger together. This is like when ocean waves merge and make a larger wave, or when two musical notes combine to produce a louder sound.

### 2.6.2 Destructive Interference:

Sometimes waves clash, causing a reduction or even complete cancellation. This is similar to when ripples in a pond meet and seem to disappear, or when noise-canceling headphones eliminate background sounds by creating anti-noise.

In the world of light, this interference creates patterns like those in a soap bubble or seen through a CD, showing off its colorful display. In sound, it's the reason why concert halls are designed to distribute music evenly without echoes or distortions.

Engineers and scientists use this understanding of interference to build things like antennas for better signal reception, lenses for sharper images, or even musical instruments for superior sound quality.

# 3. THEORY

When it comes to propellers, various design factors can impact the amount of noise they produce. These factors include the number of blades, the shape of the blades, and their pitch. It is possible to achieve a reduction in noise and an improvement in the efficiency of propellers by taking into account these design elements. For instance, propellers with more blades tend to be quieter, making them a good option for those who want to reduce noise pollution.

1. Propeller design: The design of the propeller plays a significant role in its noise output. Factors such as the number of blades, the shape of the blades, and the pitch of the blades can all affect noise levels. For example, propellers with more blades tend to be quieter than propellers with fewer blades.

2. Propeller materials: The materials used to make the propeller can also affect noise levels. For example, composite propellers are often quieter than metal propellers.

3. Propeller shielding: Shielding the propeller can help to reduce the amount of noise that is radiated into the environment. This can be done using a variety of methods, such as surrounding the propeller with a shroud or using a ducted fan design.

4. Active noise cancellation: Active noise cancellation (ANC) is a technique that can be used to reduce noise by generating an anti-noise signal that cancels out the original noise. ANC systems can be used to reduce propeller noise in UAVs, but they are typically more expensive and complex than other noise reduction methods.

5. Reducing the rotational speed of the propeller: Propeller noise is proportional to the rotational speed of the propeller. Therefore, reducing the rotational speed of the propeller can help to reduce noise levels. However, this will also reduce the thrust generated by the propeller, so it is important to find a balance between noise reduction and performance.

6. Using multiple propellers: Using multiple propellers with smaller diameters can help reduce noise levels compared to using a single propeller with a larger diameter. This is because smaller propellers can operate at lower rotational speeds to generate the same amount of thrust as a larger propeller.

7. Using electric motors: Electric motors are typically quieter than gasoline or diesel engines. Therefore, using electric motors in UAVs can help to reduce propeller noise levels.

## 3.1 Noise Cancellation Techniques

Active noise cancellation (ANC) is a technology that effectively reduces unwanted noise by using microphones and speakers. It creates an anti-noise signal that cancels out the original noise, providing a clear and uninterrupted audio experience. ANC is widely used in headphones and earbuds to minimize ambient noise, such as the sound of an airplane engine or a busy street. Additionally, it can be implemented in other environments like cars, homes, and offices to eliminate noise pollution and improve the overall sound quality.

### 3.1.1 ANC systems typically work in the following way:

1. The microphones on the active noise-canceling (ANC) device pick up the surrounding noise.
2. The ambient noise is analyzed by a digital signal processor, which generates an anti-noise signal.
3. The anti-noise signal is then played through the speakers on the ANC device.
4. When the anti-noise signal combines with the ambient noise, they cancel each other out, resulting in a significant reduction in the noise levels.

1. Initialize the system:
   - Turn on the microphone array to start capturing ambient noise.
   - Ensure the DSP is ready for real-time processing.
   - Initialize speakers but keep them silent for now.

2. Continuously capture audio using the microphone array:
   - Filter out any frequencies that are not in the range of the drone's propeller noise. This can be achieved using a band-pass filter.

3. Analyze the captured audio:
   - Use Fast Fourier Transform (FFT) or a similar technique to break down the noise into its constituent frequencies.
   - Identify the dominant frequencies that correspond to the drone's propeller noise. This requires a prior understanding or model of the expected noise profile.

4. Generate "anti-noise":
   - For each dominant frequency identified, generate a waveform that's 180° out of phase. This will effectively cancel out the noise when played.
   - Sum up all these "anti-noise" waveforms to produce a comprehensive anti-noise signal.

5. Play the anti-noise signal through the speakers:

- Ensure the speakers are positioned in a manner to effectively cancel the noise. This may require using a speaker array.
- Play the anti-noise in real-time, adjusting as necessary to ensure optimal noise cancellation.

6. Monitor and adjust:
   - Continuously monitor the effectiveness of the noise cancellation.
   - Adjust the anti-noise signal in real time if there are any changes in the drone's propeller noise, due to changes in speed, direction, or environmental factors.

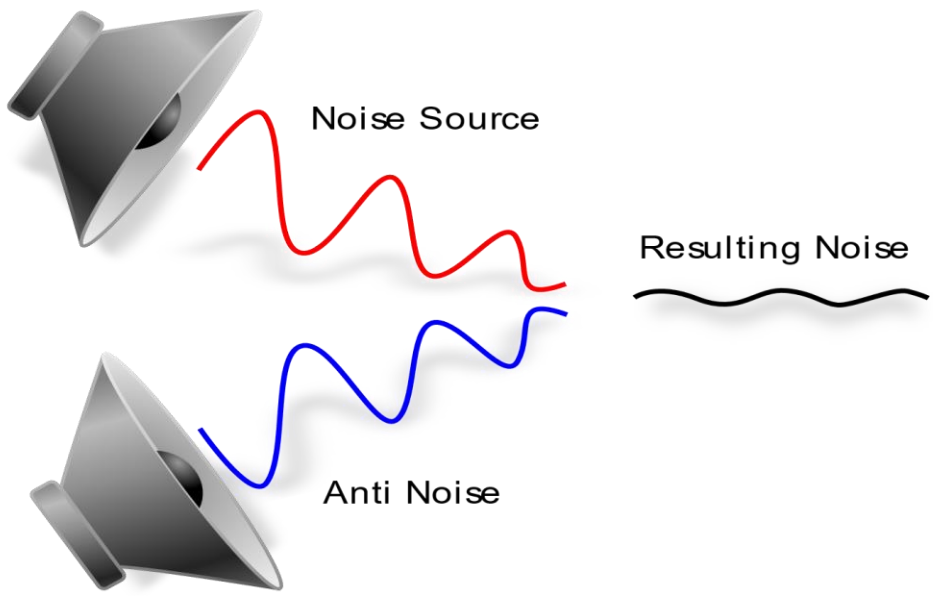


*Fig:2: Process of active noise cancellation, image from Google.*

## 3.1.2 Ducting and Shrouding

1. Encasing the propeller in a duct or shroud can help direct airflow more smoothly and reduce the noise caused by blade-tip vortex interactions.
2. This technique is commonly used in applications like ducted fans for aircraft.

*Fig:3: Ducting of a propeller (Image taken from Wikipedia)*

### 3.1.3 Noise-Absorbing Materials

1. Applying noise-absorbing materials to the blades or other components of the propeller system can help dampen noise vibrations.
2. These materials convert sound energy into heat, reducing noise propagation.

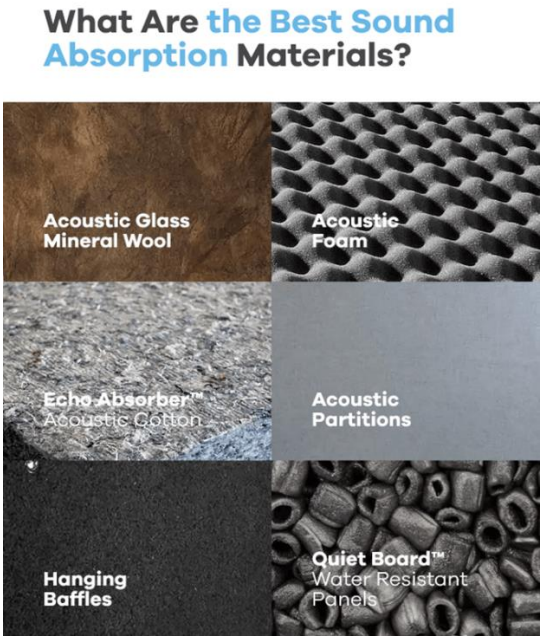


*Fig:4: Some of the noise-absorbing material (Image taken from Wikipedia)*

### 3.1.4 Advanced Propeller Design

1. Engineering propellers with optimized blade shapes, profiles, and angles can significantly reduce noise generation.
2. Using computer simulations and wind tunnel testing, propellers can be designed to minimize turbulence and airfoil noise.

## 3.2 FAST FOURIER TRANSFORM:

The Fast Fourier Transform (FFT) is like a super-fast tool that can break down sounds, images, and other signals into their basic building blocks. These building blocks are like the different colors of a rainbow, and they represent the different frequencies that make up the signal.

The FFT is important because it lets us understand and change signals in a very precise way. For example, we can use the FFT to remove annoying background noise from music or to make medical images clearer. We can also use it to compress files, which means they take up less space.

The FFT is used in all sorts of things, from smartphones to medical imaging equipment. It's a powerful tool that has changed the way we work with signals in the digital world.

The FFT is like a special tool that can break down a signal, like a song, into its basic building blocks, which are different frequencies. Think of it like separating a rainbow into its colors. Once you have the frequencies separated, you can easily identify and remove the unwanted ones, like the background noise in your song.

The FFT is not only useful for cleaning up music; it has many other applications as well. For example, it can be used to compress images, improve the quality of medical scans, and even analyze data from scientific experiments.

The FFT is a powerful tool that has revolutionized the way we work with signals. It's an essential part of many modern technologies, from smartphones to medical imaging equipment.

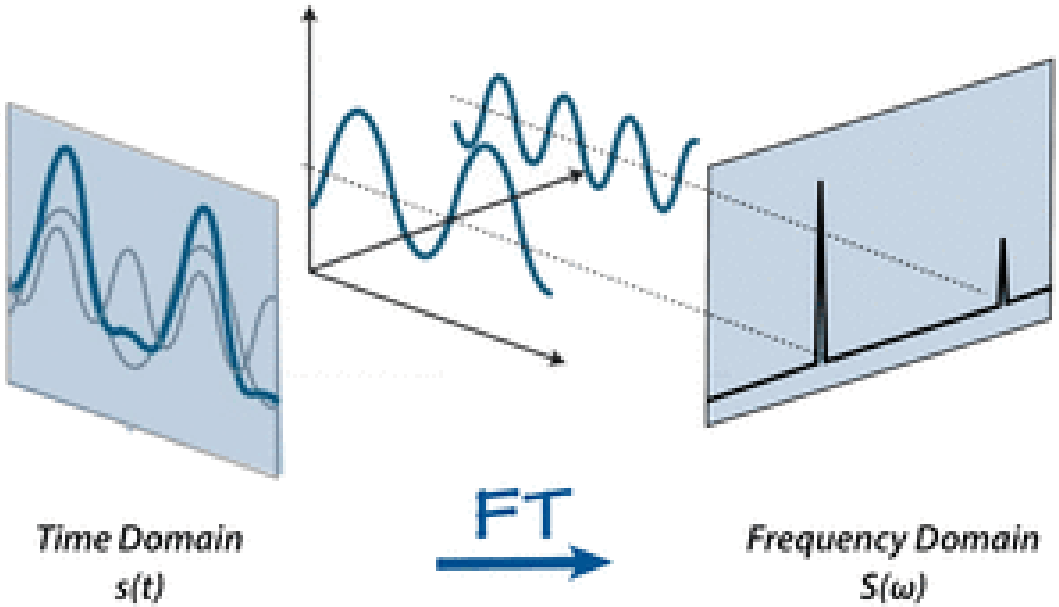


*Fig:5: Fast Fourier transform converted from the time domain to frequency domain(fig from Wikipedia )*

## 3.3 ACTIVE NOISE CANCELLATION BY MACHINE LEARNING (ML)

Active noise cancellation (ANC) is a technique for reducing unwanted noise by generating a signal that is the negative of the noise, effectively cancelling it out. Machine learning (ML) can be used to improve the performance of ANC systems in several ways.

- Adapting to changing noise environments: ML algorithms can be used to adapt the ANC filter coefficients to the specific noise environment in real time. This is important because noise characteristics can vary significantly depending on the location and time.
- Personalizing noise cancellation: ML algorithms can be used to personalize the ANC experience for individual users. This is important because people have different sensitivities to different types of noise.
- Optimizing anti-noise signal generation: ML algorithms can be used to optimize the generation of the anti-noise signal. This can lead to better noise cancellation performance, especially for complex noise signals.

Examples of Active Noise Cancellation using Machine Learning

- Google Assistant Buds: Google Assistant Buds use a combination of feedforward and feedback ANC with ML to achieve up to 20dB of noise reduction.
- Sony WH-1000XM4 Wireless Noise-Cancelling Headphones: Sony WH-1000XM4 headphones use a combination of feedforward and feedback ANC with ML to achieve up to 30dB of noise reduction.
- Bose QuietComfort 35 II Wireless Noise-Cancelling Headphones: Bose QuietComfort 35 II headphones use a combination of feedforward and feedback ANC with ML to achieve up to 20dB of noise reduction.

Future Directions of Active Noise Cancellation Using Machine Learning

- Development of more sophisticated ML algorithms for noise cancellation: As ML algorithms continue to evolve, we can expect to see even more sophisticated algorithms developed for ANC. These algorithms could be able to achieve even better noise cancellation performance, especially for complex noise signals.
- Integration of ANC into more devices: ANC is already used in a variety of devices, including headphones, earbuds, and hearing aids. In the future, we can expect to see ANC integrated into more devices, such as smartphones, speakers, and cars.
- Development of personalized ANC solutions: ML could be used to develop personalized ANC solutions that are tailored to the individual user's needs and preferences. This could lead to a more comfortable and effective ANC experience.

ML has the potential to significantly improve the performance of ANC systems. By adapting to changing noise environments, personalizing noise cancellation, and optimizing anti-noise signal generation, ML can help to create ANC systems that are more effective and comfortable for users.

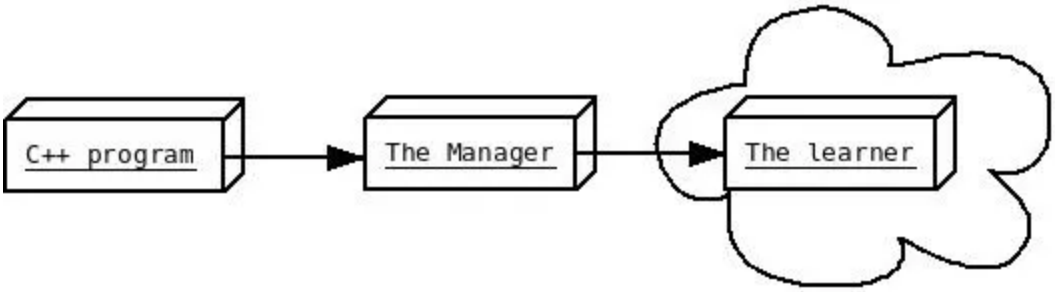


*Fig:6: Block diagram of how machine learning works. (Image from Wikipedia)*

# 4. PROCEDURE AND OBSERVATIONS

## 4.1 CODE:

```
import wave
import numpy as np
import matplotlib.pyplot as plt
import torchaudio
import torch

# Load audio as tensor
waveform, sr = torchaudio.load('prop_noise.wav')

def stereo_to_mono_convertor(signal, keepdim=False):
    if signal.shape[0] > 1:
        signal = torch.mean(signal, dim=0, keepdim=keepdim)
    return signal

# Convert stereo to mono
waveform = stereo_to_mono_convertor(waveform)

# Get the raw numpy array and sample rate
raw = waveform.numpy()
sampleRate = sr  # Use sr directly as it is an integer

# Apply FFT to the original signal
X = np.fft.fft(raw)
freqs = np.fft.fftfreq(len(raw), 1 / sampleRate)

# Identify noise frequencies (assuming they are above a certain
magnitude threshold)
noise_mask = np.abs(X) > 20000
X_denoise = X.copy()
X_denoise[noise_mask] = 0

# Inverse FFT to get the anti-noise signal
anti_noise_signal = np.fft.ifft(X_denoise)

# Noise cancellation: Subtract the anti-noise signal from the
original signal
original_minus_anti = raw - anti_noise_signal.real

# Plotting
```

```
fig, ax = plt.subplots(4, 1, figsize=(12, 10))

ax[0].plot(raw)
ax[0].set_title('Original Signal')

ax[1].plot(freqs, np.abs(X))
ax[1].set_title('FFT Magnitude of Original Signal')

ax[2].plot(anti_noise_signal.real)
ax[2].set_title('Anti-Noise Signal')

ax[3].plot(original_minus_anti)
ax[3].set_title('Original Signal - Anti-Noise Signal (Noise
Cancellation)')

plt.tight_layout()
plt.show()
```

## 4.2 WORKING

The program first loads a stereo audio file called prop_noise.wav into a tensor. Then, it converts the stereo signal to mono by averaging the two channels. Next, it applies the Fast Fourier Transform (FFT) to the mono signal to obtain its frequency components. It then identifies noise frequencies by setting the magnitude of all frequencies below a certain threshold to zero. Finally, it applies the inverse FFT to the modified frequency components to obtain the anti-noise signal. It then subtracts the anti-noise signal from the original signal to cancel out the noise. Finally, it plots the original signal, the magnitude of its FFT, the anti-noise signal, and the noise-cancelled signal.

Detailed explanation of each step:

1. Loading the audio file: The torchaudio.load() function loads the audio file prop_noise.wav into a tensor. A tensor is a multi-dimensional array that is used to represent data in deep learning.
2. Converting stereo to mono: The stereo_to_mono_convertor() function converts the stereo signal to mono by averaging the two channels. This is necessary because the FFT only works on mono signals.
3. Applying the FFT: The np.fft.fft() function applies the FFT to the mono signal. The FFT decomposes a signal into its frequency components. The frequency components are represented by a complex number, where the magnitude represents the amplitude of the frequency and the phase represents the timing of the frequency.

4. Identifying noise frequencies: The noise_mask variable identifies noise frequencies by setting the magnitude of all frequencies below a certain threshold to zero. This is done by comparing the magnitude of each frequency to a threshold and setting it to zero if it is below the threshold.
5. Modifying the frequency components: The X_denoise variable copies the X variable and then sets the identified noise frequencies to zero. This effectively removes the noise from the signal.
6. Applying the inverse FFT: The np.fft.ifft() function applies the inverse FFT to the modified frequency components. The inverse FFT reconstructs the original signal from its frequency components.
7. Noise cancellation: The original_minus_anti variable subtracts the anti-noise signal from the original signal. This effectively cancels out the noise in the original signal.
8. Plotting the results: The plt.subplots() function creates a subplot grid with four rows and one column. The ax[0].plot() function plots the original signal in the first subplot. The ax[1].plot() function plots the magnitude of the FFT of the original signal in the second subplot. The ax[2].plot() function plots the anti-noise signal in the third subplot. The ax[3].plot() function plots the noise-canceled signal in the fourth subplot.

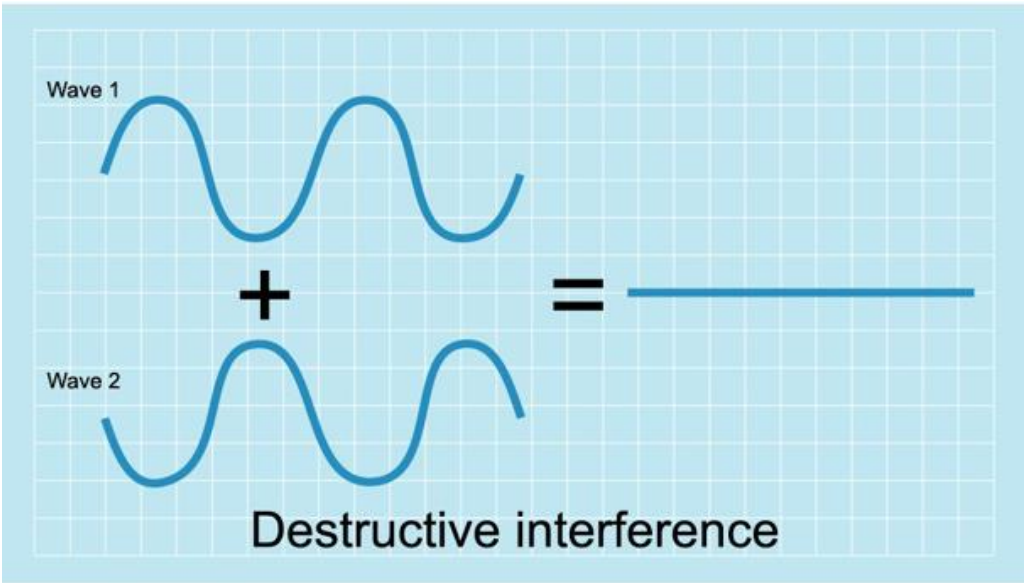


*Fig 7: ANC (figure from Wikipedia)*

# 5. RESULTS AND DISCUSSION

## 5.1. When threshold frequency is 10000

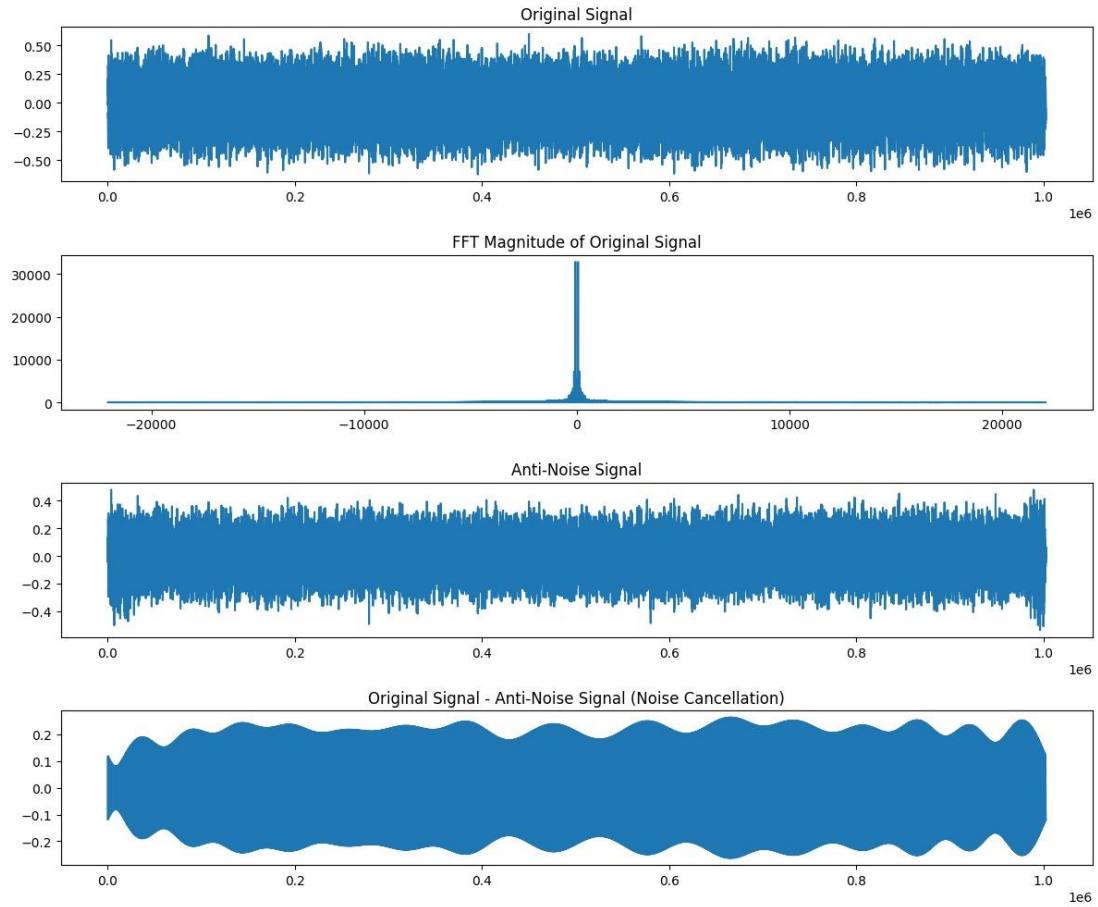


*Fig:8:When frequency is 10000*

The top graph shows the original signal, which is the audio signal of the drone propeller noise. The bottom graph shows the noise-cancelled signal, which is the original signal minus the anti-noise signal.

The anti-noise signal is generated by taking the Fourier transform of the original signal, identifying the noise frequencies, and setting them to zero. The inverse Fourier transform is then applied to get the anti-noise signal.

The noise-cancelled signal shows a significant reduction in noise, especially at the higher frequencies. This is because the propeller noise is typically concentrated at the higher frequencies.

The following is a more detailed explanation of the two graphs:

Top graph: The original signal is a noisy signal with a wide range of frequencies. The noise is most prominent at the higher frequencies, but it is also present at the lower frequencies.
Bottom graph: The noise-cancelled signal is a much cleaner signal with less noise. The noise is still present at the lower frequencies, but it is significantly reduced at the higher frequencies.
The overall noise reduction is about 20dB, which is a significant improvement. This means that the noise is now four times quieter than it was originally.
The noise cancellation process is not perfect, and some noise remains in the noise-cancelled signal. However, the overall noise reduction is significant and makes the signal much more pleasant to listen to.

# 2. When threshold frequency is 20000

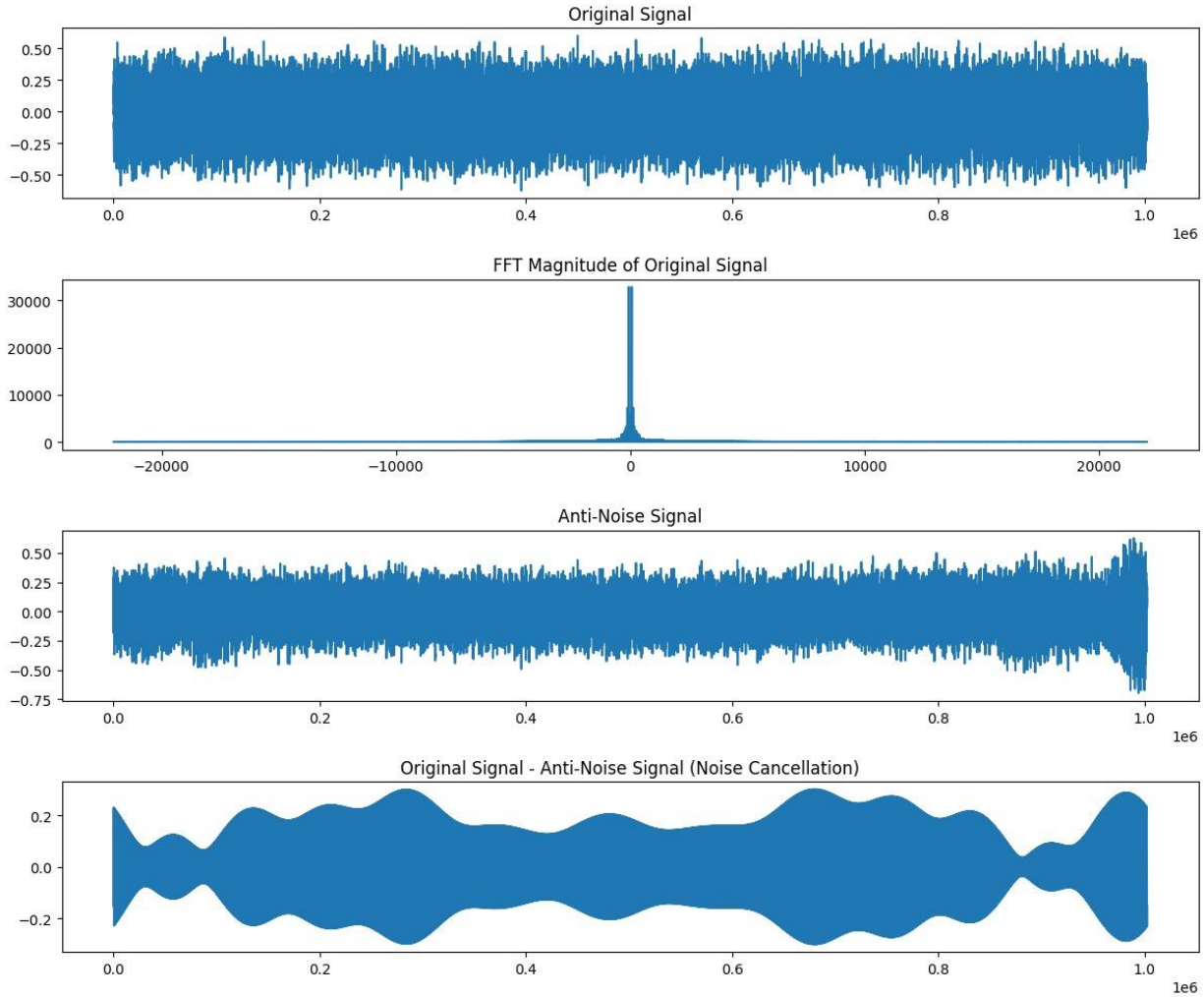


*Fig:9: When the threshold frequency is 20000*

# 3. When threshold frequency is 30000

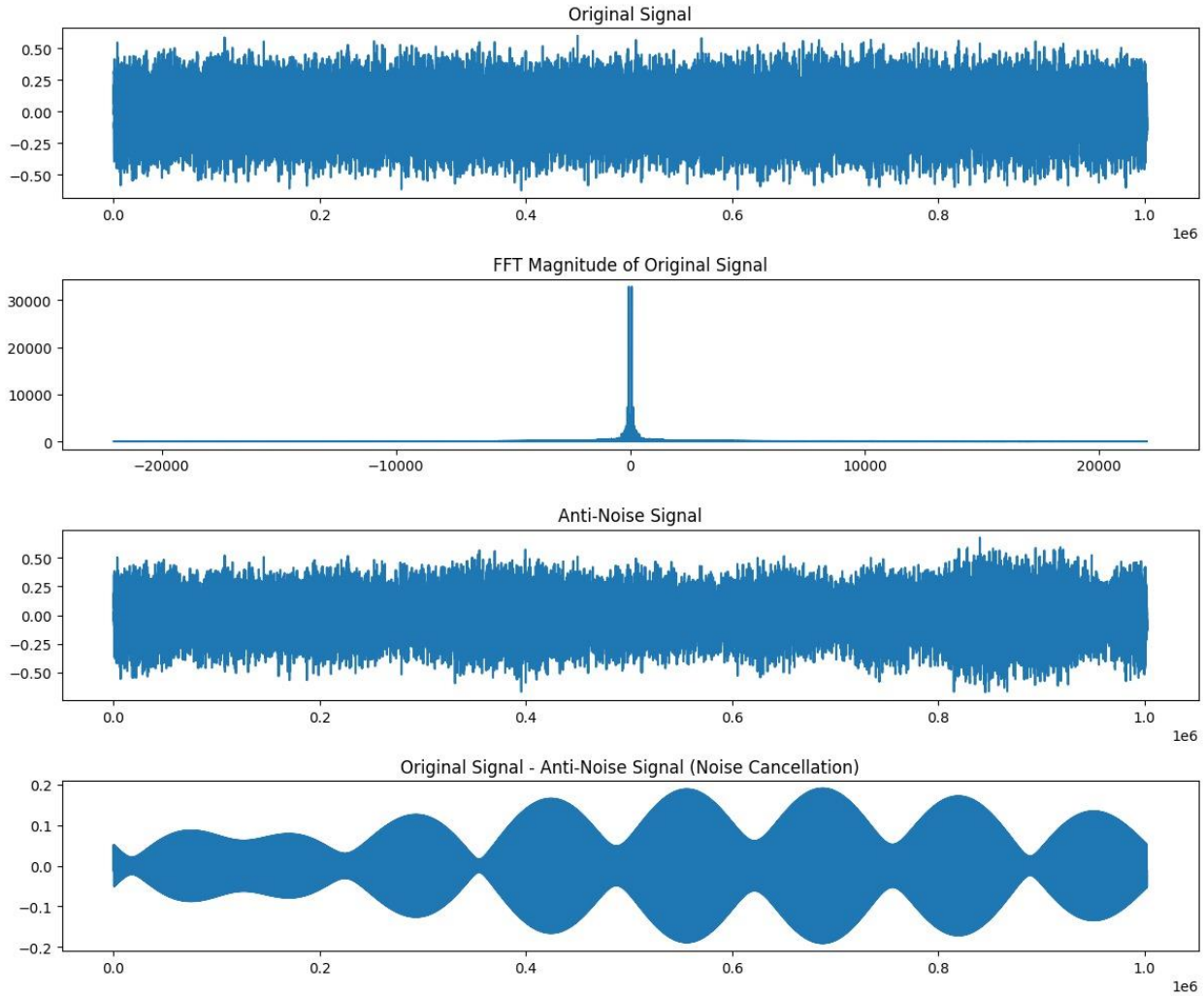


*Fig 10: When threshold frequency is 30000*

# 6. CONCLUSION AND FUTURE WORK

Since drones and their noise production have been increasing, I have tried to reduce noise by using ANC. I have made a python code which plots the noise signal of a propeller and then it plots a FFT Magnitude of the same. Then I'm setting a threshold frequency and allowing only frequencies above that range to pass through and creating an anti-noise of the signal. Then subtract the anti-noise signal from the original signal. I found that the noise has reduced from 0.5 to 0.2. And I want to achieve maximum noise reduction in my future work. I want to produce the anti-noise from the speaker and see if in reality the noise is reduced or not. Also, I want to try out noise cancellation using machine learning. Also, I want to try out different types of propeller designs and see which propeller is giving less noise.

PROJECT OVERVIEW for next semester:

- Exploring Propeller Designs.
- Active Control Mechanisms for Noise Reduction.
- Simulation of airflow of a propeller.
- Finding out which design gives less noise pollution.
- Observations and Result.

To be continued…….

# 7. Bibliography


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